

FinSphere AI Super App

Database Data Model & Relational Schema Documentation

Introduction

This document details the database schema, entity structures, attributes, key constraints, and relationships powering the FinSphere AI Super App. The backend architecture runs on Node.js/Express connected to PostgreSQL via Prisma ORM.

User Model

User account registry holding user profiles, authentication credentials, settings, and baseline monthly salaries.

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique identifier for the account
name	String	-	-	User's full name
email	String	UK	Unique index	Registered email used for logins
passwordHash	String	-	-	Securely salted and hashed password
monthlyIncome	Decimal	-	Default: 150000	Baseline monthly salary parameter
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

Expense Model

Individual transaction entries classifying cash flow (linked 1-to-many with User). Supports the sankey diagram and ledger views.

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique transaction identifier
userId	String	FK	User.id (Cascade)	Owner of the transaction
amount	Decimal	-	Positive value	Monetary value of transaction
category	Enum	-	ExpenseCategory	Category classification tag
description	String	-	-	Merchant or description notes
date	DateTime	-	-	Transaction occurrence date
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

Budget Model

Monthly category limits defined by the user (unique per user, category, and target month).

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique budget identifier
userId	String	FK	User.id (Cascade)	Owner of the budget constraint
category	Enum	-	ExpenseCategory	Target budgeted category
limitAmount	Decimal	-	Positive value	Allowed budget threshold limit
month	String	-	e.g. '2026-06'	Target calendar month
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

Goal Model

Wealth planning target savings goals detailing target date, target amount, and current progress.

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique goal identifier
userId	String	FK	User.id (Cascade)	Owner of the goal
title	String	-	-	Goal description (e.g. 'Car downpayment')
targetAmount	Decimal	-	-	Financial savings milestone target
currentAmount	Decimal	-	-	Amount accumulated so far
targetDate	DateTime	-	-	Expected completion target date
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

Investment Model

Asset holdings representing stocks, mutual funds, gold, crypto, and systematic investment plans (SIPs).

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique holding identifier
userId	String	FK	User.id (Cascade)	Owner of the asset holding
assetType	Enum	-	InvestmentType	Classification enum of the asset type
name	String	-	-	Asset name (e.g. 'Tesla Stock')
investedAmount	Decimal	-	-	Principal investment amount
currentValue	Decimal	-	-	Current valuation of asset
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

CreditProfile Model

Credit rating metrics including credit utilization, payment history, and credit age (linked 1-to-1 with User).

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique credit profile identifier
userId	String	FK, UK	User.id (Cascade, 1:1)	Associated user (unique constraint)
utilization	Decimal	-	Percentage scale	Current revolving credit utilization ratio
paymentHistory	Decimal	-	Percentage scale	Ratio of on-time payments made
creditAge	Decimal	-	Years count	Calculated average age of accounts
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

FinancialHealth Model

Ratings generated by the AI advisor engine containing score metrics (linked 1-to-1 with User).

Field Name	Type	Key	Constraints	Description
id	String	PK	UUID default	Unique financial report identifier
userId	String	FK, UK	User.id (Cascade, 1:1)	Associated user (unique constraint)
score	Int	-	Range 0 - 100	Overall health rating index
grade	String	-	e.g. 'Excellent'	Grade string label representation
createdAt	DateTime	-	@default(now())	Record creation timestamp
updatedAt	DateTime	-	@updatedAt	Last modification timestamp

System Enums

Enum Type	Values
ExpenseCategory	Food, Travel, Shopping, Bills, Entertainment, Education, Health, Other
InvestmentType	Stock, MutualFund, SIP, Gold, Crypto, Other